Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE
Course Code & Title: EE8403 Measurements and Instrumentation
Name of the Faculty member: Mrs. G. Sivapriya, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit V / Smart sensors
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 25
- **Activity Chosen: Flipped Class Room**
- **Justification:**
    - Flipped classroom is a “pedagogical approach in which direct instruction moves from the group learning space to the individual learning space, and the resulting group space is transformed into a dynamic, interactive learning environment where the educator guides students as they apply concepts and engage creatively in the subject matter”

- **Time Allotted for the Activity:** 30 Minutes

Total Strength is 30

Reporter: Myself

At the Class (30 minutes)

✓ I shared the material related to the topic through canvas LMS.
✓ Then I told them to prepare the topic as a team with a help of PPT.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO9	PO10	PO12	PSO1	PSO2	PSO3
C212.3	3	2	1	1	1	1	1	1

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO9**	**PO10**	**PO12**	**PSO1**	**PSO2**	**PSO3**
	3	**2**	**1**	**1**	**1**	**1**	**1**	**1**
Justification for correlation	Students will apply the knowledge of Electrical and Electronics engineering fundamental concepts to understand the working of transducers.	Students will compare and contrast alternative solution processes to select the best process for the specification of transducers and sensors.	Students will present results as a team, with smooth integration of contributions from all individual efforts through Active Learning Methods.	Students will be able to create flow in a document or presentation in the technical concept through Active Learning Methods.	Students will describe the rationale for the requirement for continuing professional development.	Students will be able to analyze the performance of electrical and electronic systems using modern software packages.	Students will be able to design and test the electronic system in the engineering applications	Students will be able to design and develop the hardware and software with measurement skills required for industrial automation systems.

- **Images / Screenshot of the practice:**


Benefits:

- This practice improves the peer coaching / studying with in the class. Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

References:

1. https://www.panopto.com/blog/7-unique-flipped-classroom-models-right/
2. A.K. Sawhney, 'A Course in Electrical & Electronic Measurements & Instrumentation', Dhanpat Rai and Co, 2010.

Signature of Faculty Member **HOD**

Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE-B
Course Code & Title: EE8403 Measurements and Instrumentation
Name of the Faculty member: Ms. G. Sivapriya, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit I / Principle and types of analog voltmeters and ammeters
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 6
- **Activity Chosen: Demonstration**
- **Justification:**
  - Explain the principle and types of analog meter
  - The topic Principle and types of analog meter has 7 types of instruments. Since each type has its own principle of operation, torque equation and construction. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the each type of meters which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

After teaching the concept, give students one or two minutes to think about the topic without writing anything.
Total Strength is 29,
Reporter: Myself

- At the end the Class (Last 5 minutes)
- I asked the students to think about various types of meters in measurements concept for 2 minutes.
- Then I told them to I told them to Pair with their neighbours and discuss about the construction parts of instruments for another 1 minute.
- Finally, I show the PMMC and PMMI ammeter for each row (3 minutes)

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PO12
C212.1	3	1	1	1	1	1

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO10**	**PO12**
	3	**1**	**1**	**1**	**1**	**1**
Justification for correlation	Students will apply the knowledge of Electrical and Electronics engineering fundamental concepts to understand the basics of measurement and instruments.	Students will identify the mathematical, engineering and other relevant knowledge that applies to an Analog and digital measurement systems.	Students will determine the design objectives and functional requirements of the instruments.	Students will examine the relevant methods, tools and techniques of system calibration.	Students will be able to create flow in a document or presentation.	Students will describe the rationale for the requirement for continuing professional development.

- **Images / Screenshot of the practice:**

❖ *Reflective Critique:*

1. **Pre-implementation Reflection :**

- **Benefits:**
  - ➢ Students can able to attend the question even in the questions are in indirect form.
  - ➢ Students can able to explain the concepts in examination without any confusion.
- **Challenges:**
  - ➢ In the class mostly boys hesitate to answer the questions.
  - ➢ Time utilization for conducting activity.

2. **Post-implementation Reflection :**

- **Benefits:**
  - ➢ Students understood the concept which was reflected from their answers forthe questions I have asked during discussion session.
- **Challenges:**
  - ➢ Slow learners were not able to understand some topics during discussion hours.

❖ *Benefit of the practice***:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students can able to explain the Analog meters (MC, MI, and Electrodynamometer) construction and working principle.
- ✓ The success of the activity was evaluated by asking the same question in **Internal**

Assessment test I – Around 80% of students answered.

References:

1. A.K. Sawhney, 'A Course in Electrical & Electronic Measurements &Instrumentation', Dhanpat Rai and Co, 2010.
2. https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/172-demonstrations

Signature of Faculty Member **HOD**

Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE-B
Course Code & Title: EE8403 Measurements and Instrumentation
Name of the Faculty member: Mrs. G. Sivapriya, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit I / Errors in measurement
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 3
- **Activity Chosen: One minute paper**
- **Justification:**
  - Explain various types of errors

  - The topic Errors in Measurements has 9 types of errors in instruments. Since each type has its own definition, limitation and examples. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the each type of errors which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

After teaching the concept, give students one or two minutes to think about the topic withoutwriting anything.

Total Strength is 30

Reporter: Myself

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about various types of errors in measurements concept for 3 minutes.
- ✓ Then I told them to write as much as they can within a short period of time (1 minute)
- ✓ Finally, I collected the papers from each column (1 minute)

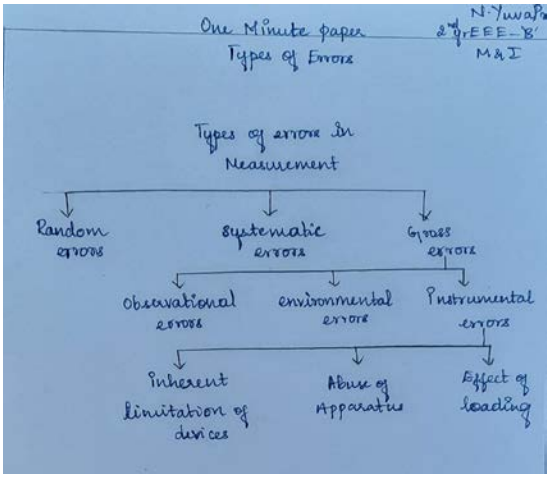
- **CO – PO / PSO mapping:**

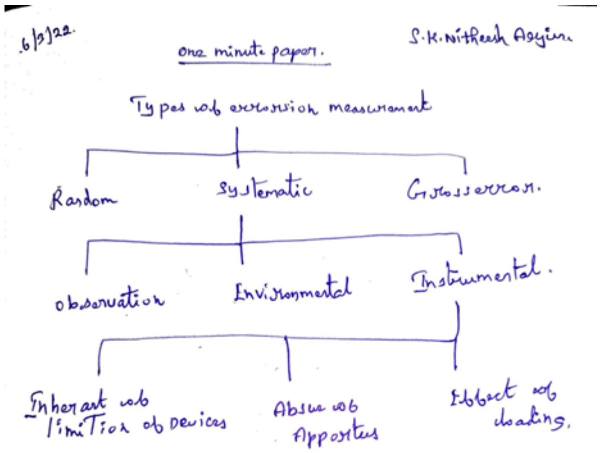
CO	PO1	PO2	PO3	PO4	PO10	PO12
C212.1	3	1	1	1	1	1

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO10	PO12
	3	1	1	1	1	1
Justification for correlation	Students will apply the knowledge of Electrical and Electronics engineering fundamental concepts to understand the basics of measurement and instruments.	Students will identify the mathematical, engineering and other relevant knowledge that applies to an Analog and digital measurement systems.	Students will determine the design objectives and functional requirements of the instruments.	Students will examine the relevant methods, tools and techniques of system calibration.	Students will be able to create flow in a document or presentation.	Students will describe the rationale for the requirement for continuing professional development.

- **Images / Screenshot of the practice:**





❖ *Reflective Critique:*

1. **Pre-implementation Reflection :**

- **Benefits:**

  ➢ Students can able to attend the question even in the questions are in indirect form.

- Students can able to explain the concepts in examination without anyconfusion.

- **Challenges:**
  - In the class mostly boys hesitate to answer the questions.
  - Time utilization for conducting activity.

2. **Post-implementation Reflection :**

- **Benefits:**
  - Students understood the concept which was reflected from their answers forthe questions I have asked during discussion session.
- **Challenges:**
  - Slow learners were not able to understand some topics during discussionhours.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explainvarious types of errors present in measurements.
- ✓ The success of the activity was evaluated by asking the same question in **InternalAssessment test I – Around 60% of students answered.**

References:

❖ https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-one-minute-paper

Signature of Faculty Member **HOD**

Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE- B
Course Code & Title: EE8403 Measurements and Instrumentation
Name of the Faculty member: Ms. G. Sivapriya, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit II / Single, three phase energy meter
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 8
- **Activity Chosen: Demonstration**
- **Justification:**
    - Explain Wattmeter and Energy meter
    - After teaching the concept, I thought of conducting this activity for making the students to give the difference between the Wattmeter and Energy meter which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

After teaching the concept, give students one or two minutes to think about the topic without writing anything.
Total Strength is 31
Reporter: Myself

At the end the Class (Last 5 minutes)

- I asked the students to think about various types of meters in measurements concept for 2 minutes.
- Then I told them to I told them to pair with their neighbours and discuss about the construction parts of instruments for another 1 minute.
- Finally, I show the Electrodynamometer type wattmeter and induction type energy meter for each row (3 minutes)

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PO12
C212.2	3	2	1	1	1	1

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO10**	**PO12**
	3	**1**	**1**	**1**	**1**	**1**
Justification for correlation	Students will apply the knowledge of Electrical and Electronics engineering fundamental concepts to understand the working of meters.	Students will compare and contrast alternative solution processes to select the best process for the measurement.	Students will determine the design objectives and functional requirements of the instruments.	Students will examine the relevant methods, tools and techniques of system calibration.	Students will be able to create flow in a document or presentation.	Students will describe the rationale for the requirement for continuing professional development

- **Images / Screenshot of the practice:**

❖ *Reflective Critique:*

1. **Pre-implementation Reflection :**
   - **Benefits:**

- Students can able to attend the question even in the questions are in indirect form.
- Students can able to explain the concepts in examination without any confusion.

- **Challenges:**
  - In the class mostly girls hesitate to answer the questions.
  - Time utilization for conducting activity.

2. **Post-implementation Reflection :**

- **Benefits:**
  - Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- **Challenges:**
  - Slow learners were not able to understand some topics during discussion hours.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The assessment of effectiveness of the activity was felt when told most of the points.

- While conducting the activity, I understood that the students can able to explain the analog meters (Electrodynamometer and Wattmeter) construction and working principle.

- The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 80%** of students answered.

References:

1. A.K. Sawhney, 'A Course in Electrical & Electronic Measurements &Instrumentation', Dhanpat Rai and Co, 2010.
2. https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/172-demonstrations

Signature of Faculty Member **HOD**

Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE
Course Code & Title: EE8403 Measurements and Instrumentation
Name of the Faculty member: Mrs. G. Sivapriya, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit III / Electrostatic and electromagnetic interference
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 15
- **Activity Chosen: Sage and scribe**
- **Justification:**
    - Differentiate Electrostatic and electromagnetic interference
    - After teaching the concept, I thought of conducting this activity for making the students to give the difference between the Electrostatic and electromagnetic interference which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

After teaching the concept, give students one or two minutes to think about the topic without writing anything and tell pair with neighbour 2 in desk.

Total Strength is 30, Number of Pairs – 15

Reporter: Myself

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about types of interference in measurements concept for 2 minutes.
- ✓ Then I told them to I told them to pair with their neighbours and play one person as sage role and other person as scribe role.
- ✓ Scribe take a notes about topic with the help of sage instruction (3 minutes)

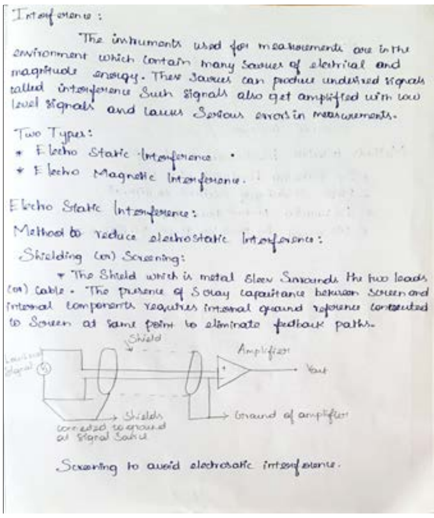
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1
C212.3	3	2	1	1	1	1	1	1

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO9**	**PO10**	**PO12**	**PSO1**
	3	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Justification for correlation	Students will apply the knowledge of Electrical and Electronics engineering fundamental concepts to understand the working of meters.	Students will compare and contrast alternative solution processes to select the best bridge for the measurement.	Students will determine the design objectives and functional requirements of the measurement bridges.	Students will examine the relevant methods, tools and techniques of system calibration	Students will demonstrate effective communication, problem-solving, conflict resolution and leadership skills with their team through Active Learning Methods.	Students will be able to create flow in a document or presentation in the technical concept through Active Learning Methods.	Students will describe the rationale for the requirement for continuing professional development.	Students will be able to analyze the performance of electrical and electronic systems.

- **Images / Screenshot of the practice:**



Interference :

The instruments used for measurements are in the environment which contain many sources of electrical and magnitude energy. These sources can produce undesired signals called interference Such signals also get amplified with low level signals and cause serious errors in measurements.

Two Types:

* Electro Static Interference.
* Electro Magnetic Interference.

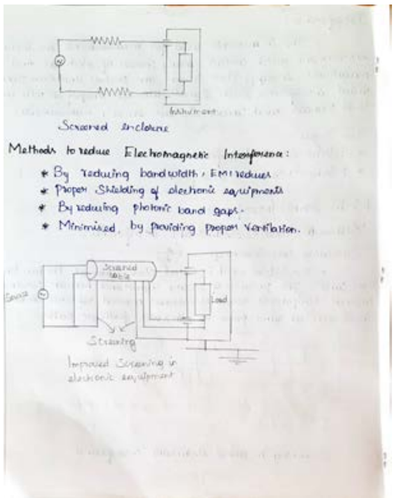
Electro Static Interference:

Method to reduce electrostatic Interference:

Shielding (or) Screening:

* The Shield which is metal sleev surrounds the two leads (or) cable. The presence of stray capacitance between screen and internal components requires internal ground reference connected to screen at same point to eliminate feedback paths.

Screening to avoid electrostatic interference.



Screened enclosure

Methods to reduce Electromagnetic Interference:

* By reducing bandwidth, EMI reduces
* Proper Shielding of electronic equipments
* By reducing photonic band gaps.
* Minimized, by providing proper ventilation.

Improved Screening in electronic equipment

❖ *Reflective Critique:*

1. **Pre-implementation Reflection :**

Benefits:

- Students can able to attend the question even in the questions are in indirect form.
- Students can able to explain the concepts in examination without any confusion.

Challenges:

- In the class mostly girls hesitate to answer the questions.
- Time utilization for conducting activity.

2. **Post-implementation Reflection :**

Benefits:

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

Challenges:

- Slow learners were not able to understand some topics during discussion hours.

- *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The assessment of effectiveness of the activity was felt when told most of the points.

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students can able to explain the difference between Electrostatic and electromagnetic interference
- ✓ The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 60% of students answered.

References:

1. A.K. Sawhney, 'A Course in Electrical & Electronic Measurements & Instrumentation', Dhanpat Rai and Co, 2010.
2. https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/172-Sage and scribe

Signature of Faculty Member **HOD**

Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE-B
Course Code & Title: EE8403 Measurements and Instrumentation
Name of the Faculty member: Ms. G. Sivapriya, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit IV / Cathode Ray Oscilloscope
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 19
- **Activity Chosen: Think – Pair - Share**
- **Justification:**
    - After teaching the concept, I thought of conducting this activity for making the students able to explain the parts of CRT display which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes
After teaching the concept, the students were made to pair with their neighbors

Photographer: Myself

Reporter: Myself

At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about the parts of CRT display for 2 minute.
- ✓ Then I told them to Pair with their neighbours and discuss about the concepts for another 1 minute.
- ✓ Finally I told them to share the points and draw the design within 3 minutes.

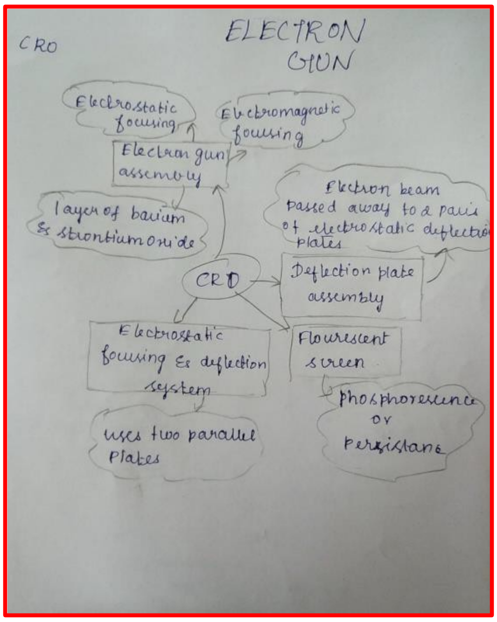
- **CO – PO / PSO mapping:**

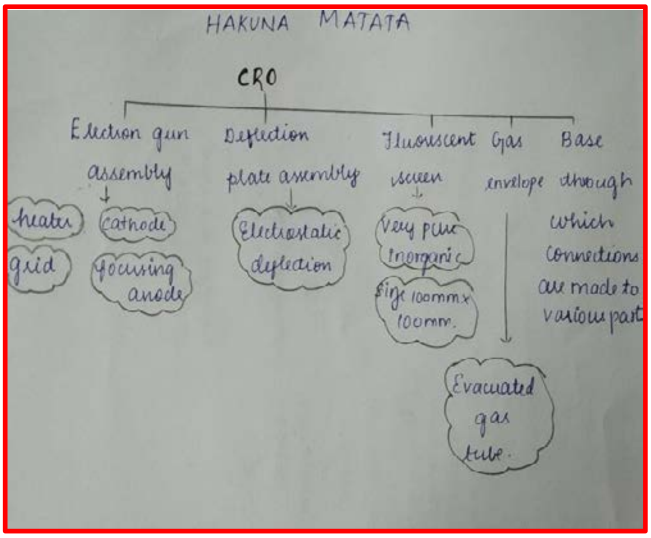
CO	PO1	PO2	PO9	PO10	PO12	PSO2
C212.4	3	2	1	1	1	1

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO9	PO10	PO12	PSO2
	3	1	1	1	1	1
Justification for correlation	Students will apply the knowledge of Electrical and Electronics engineering fundamental concepts to understand the working of display and printing devices.	Students will compare and contrast alternative solution processes to select the best measurement method of various display instruments.	Students will examine the relevant methods, tools and techniques of data acquisition systems.	Students will be able to create flow in a document or presentation in the technical concept through Collaborative Learning Methods.	Students will describe the rationale for the requirement for continuing professional development.	Students will be able to design and test the electronic system in the engineering applications.

- **Images / Screenshot of the practice:**





❖ *Reflective Critique:*

1. **Pre-implementation Reflection :**

Benefits:

- ➢ Students can able to attend the question even in the questions are in indirect form.
- ➢ Students can able to explain the concepts in examination without any

confusion.

Challenges:

- ➢ In the class mostly girls hesitate to answer the questions.
- ➢ Time utilization for conducting activity.

2. Post-implementation Reflection :

Benefits:

- ➢ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

Challenges:

- ➢ Slow learners were not able to understand some topics during discussion hours.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The assessment of effectiveness of the activity was felt when told most of the points.

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students can able to explain the parts of CRO.
- ✓ The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 60% of students answered.

References:

1. A.K. Sawhney, 'A Course in Electrical & Electronic Measurements & Instrumentation', Dhanpat Rai and Co, 2010.
2. https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/172

Signature of Faculty Member **HOD**